

Luís Pedro Borges

Machine Learning | Information Retrieval | Neural Search

Lisbon, Portugal | +351 935 922 727 | lpborges.careers@gmail.com

luispb7.github.io | GitHub: LuisPB7 | LinkedIn | Google Scholar

Canada: IEC Working Holiday approved for up to 2 years.

Open work permit activation on arrival; no employer sponsorship required during the permit period.

Profile

Machine learning and information retrieval researcher with a **Ph.D. in progress** at Instituto Superior Técnico and Carnegie Mellon University. Experience in neural retrieval, knowledge distillation and LLM re-ranking, including benchmark experiments at scale and search efficiency evaluation. Seeking Research Scientist, Data Scientist, Data Engineer, ML Engineer and Search Engineer roles and internships in Canada.

Technical Skills

Programming: Python, C++, Java, C.

Machine learning: PyTorch, TensorFlow, Keras, NumPy, SciPy, scikit-learn.

Search and NLP: learned sparse and dense retrieval, inverted indexes, PISA, passage re-ranking, knowledge distillation, LLM re-ranking, named entity recognition, fact verification.

Evaluation: MS MARCO, TREC Deep Learning and BEIR; ranking quality, recall and retrieval latency.

Research Experience

Early Stage Researcher

Mar 2025 - Present

INESC-ID, Lisbon, Portugal

- Research knowledge distillation methods for small language models in information retrieval.

Doctoral Research - Ph.D. in progress

Sep 2019 - Present

Instituto Superior Técnico / INESC-ID and Carnegie Mellon University

- Developed **KALE**, connecting learned sparse query and document representations with efficient inverted index search; evaluated on MS MARCO, TREC and BEIR.
- Conducted PyTorch distillation experiments using approximately **45 million existing synthetic query-passage judgments**, with training on a single NVIDIA A100 GPU.
- Developed and evaluated **EmberTiny**, a retriever with 14.5M parameters that retained **97% of the MRR@10 of its SimLM teacher with 110M parameters** on MS MARCOv1 (0.403 versus 0.415).
- Developed **SCALE** and evaluated it with **SPLADE.Lex**: **44% lower mean index search latency** than SPLADEv3 (60 versus 108 ms), with MRR@10 improving from 0.400 to 0.415 in MS MARCOv1 dissertation experiments.
- Developed retrieval and zero-shot LLM re-ranking for Tip-of-the-Tongue search across domains; published at SIGIR 2024, with public code and data.

Additional Experience

Teaching Assistant - Search Engines

Jan 2022 - May 2022

Carnegie Mellon University, Pittsburgh, United States

- Supported the Search Engines course under Professor Jamie Callan.

Junior Researcher

Mar 2019 - Sep 2019; Jul 2018 - Dec 2018

INESC-ID, Lisbon, Portugal

- Worked on string matching tasks in natural language processing across two research appointments.

Visiting Junior Researcher

Jan 2019 - Apr 2019

National Institute of Informatics, Tokyo, Japan

- Worked on the FEVER fact verification task.

Junior Researcher Intern

Jul 2018 - Sep 2018

Priberam, Lisbon, Portugal

- Worked on the OntoNotes 5.0 named entity recognition task.

Education

Ph.D. in Language and Information Technologies - in progress

2019 - Present

Instituto Superior Técnico and Carnegie Mellon University

Dual degree programme. Advisors: Bruno Martins (IST) and Jamie Callan (CMU).

Dissertation research: *Latent Vocabularies for Learned Sparse Retrieval*.

M.Sc. in Information Systems and Computer Engineering

2016 - 2018

Instituto Superior Técnico, University of Lisbon

Thesis: *Detecting Fake News Using Deep Neural Networks*.

B.Sc. in Information Systems and Computer Engineering

2013 - 2016

Instituto Superior Técnico, University of Lisbon

Selected Publications

Generalizable Tip-of-the-Tongue Retrieval with LLM Re-ranking

L. Borges, R. Jha, J. Callan and B. Martins. *SIGIR*, 2024.

Paper

KALE: Using a K-Sparse Projector for Lexical Expansion

L. Borges, B. Martins and J. Callan. *ICTIR*, 2023.

Paper

Assessing the Benefits of Model Ensembles in Neural Re-ranking for Passage Retrieval

L. Borges, B. Martins and J. Callan. *ECIR*, 2021.

Paper

Full publication list and open papers